

Jonathan Navarro

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EDUCATION

Bachelor of Science in Computer Science The University of Texas Rio Grande Valley, Edinburg, TX
May 2026

Magna Cum Laude | **Dean's List** (2023-2024) | **President's List** (2024-2026)

CS EXPERIENCE

Rise Above Pressure Program Website <https://rapp956-website.pages.dev/> 2026

- **Designed** and developed a professional **marketing** website promoting a youth mental health and life skills initiative for Rio Grande Valley school districts, creating an engaging **user experience** focused on accessibility, outreach, and **community impact**.
- **Collaborated** on branding, content presentation, and user-focused design to effectively communicate program goals, increase visibility, and support **real-world** audience engagement.

Pokémon Battle Bot – Reinforcement Learning AI 2026

- **Developed** a PPO-based **reinforcement learning** agent for autonomous Pokémon Showdown battles, implementing a custom **Gymnasium** environment with reward shaping and action masking to optimize **training efficiency**
- **Optimized** agent performance through iterative environment design and hyperparameter **fine-tuning**; monitored metrics via **TensorBoard**

HACKATHON ACHIEVEMENTS

1st Place – UTRGV Frontera Gemini Hackathon | Financial Analytics Dashboard 2024

- **Integrated Gemini** API to power intelligent spending insights within a robust **full-stack** architecture, delivering a complete, production-ready solution in under 24 hours
- **Architected** dynamic, full-stack dashboards with RESTful API integration, enabling real-time data ingestion, aggregation, and visualization of business spending and savings trends

2nd Place – RGV Tutor | Offline AI Learning App | AI/ML Track 2026

- **Solved** a real connectivity gap: an **offline-first architecture** that brought **AI** tutoring to low-connectivity environments with zero reliance on external APIs / tokens.
- **Embedded** a local LLM (Ollama) to power adaptive quizzes, flashcards, and a digital Book Hub, optimized for low-end devices in **resource-constrained** environments

Hack Research – Mechanical Computation with Roons | Research Paper 2025

- **Co-authored** research **validating** that marble-based mechanical systems can simulate a Turing Machine, contributing equally across **computational modeling**, **memory/state-transition design**, and **technical documentation** demonstrating how physical systems implement memory management, sequential logic, and algorithmic processing

EXPERIENCE

Food Delivery Driver | DoorDash 2023 – Present

- **Efficiently** managed high-volume deliveries from 100+ local restaurants as a **Gold Status driver**, demonstrating strong **time management**, customer service, problem-solving, and **communication skills** while maintaining delivery accuracy and **customer satisfaction**.

Drywall Finisher | Navarro's Drywall 2023

- **Coordinated** drywall installation and finishing for residential and **commercial projects**, maintaining **precision**, safety compliance, and **quality standards** under tight deadlines

TECHNICAL SKILLS

Machine Learning & AI: Reinforcement Learning (PPO), Stable-Baselines 3, Gymnasium, TensorFlow, PyTorch, LLM Integration, Prompt Engineering, Reward Shaping, TensorBoard **Web & Software Development:** React, Node.js, Express, MongoDB, SQLite, REST APIs, Responsive Design, Authentication (OAuth/JWT), State Management **Game Development:** Unity, 2D/3D Game Mechanics, Physics Simulation, UI Systems, Performance Optimization, Debugging & Profiling **Systems & Tools:** Linux/UNIX, AWS, Azure, Docker, Git/GitHub, CI/CD Pipelines, Bash/Shell Scripting, Cloud Deployment, Virtual Machines, VS Code **Design & Engineering:** Figma, UX/UI Design, System Architecture, Design Systems, Data Structures & Algorithms, Software Design Patterns, Testing (Unit/Integration), Scalability Analysis **Additional:** Cybersecurity, Networking, API Design, Technical Documentation, Agile Development, Project Management, Communication, Team Collaboration